SQL Data Analysis Internship – Task 1

Student Management Database — Full Report with Code, Output & Explanation

1. Overview

This report documents the design and implementation of a Student Management Database built in SQL. For every task, the actual SQL code is shown, followed by the real query output (executed and verified), and a short explanation of what the result means from a data-analysis perspective.

2. Database & Table Setup

```
CREATE DATABASE IF NOT EXISTS StudentManagement;
USE StudentManagement;

CREATE TABLE Students (
    StudentID     INT PRIMARY KEY AUTO_INCREMENT,
    Name          VARCHAR(50),
    Gender        CHAR(1),
    Age           INT,
    Grade         VARCHAR(2),
    MathScore     INT,
    ScienceScore  INT,
    EnglishScore  INT
);
```

The Students table stores one row per student, with StudentID as an auto-incrementing primary key and separate integer columns for each subject score, enabling per-subject aggregation later.

3. Insert Statements (12 Records)

```
INSERT INTO Students (Name, Gender, Age, Grade, MathScore, ScienceScore, EnglishScore) VALUES
('Aarav Sharma',  'M', 15, '9A',  85, 78, 88),
('Priya Nair',    'F', 16, '10A', 92, 95, 89),
('Rohan Mehta',   'M', 15, '9B',  67, 72, 74),
('Sneha Kapoor',  'F', 16, '10B', 78, 82, 91),
('Karan Verma',   'M', 14, '9A',  55, 60, 65),
('Ananya Iyer',   'F', 15, '9B',  95, 91, 93),
('Vikram Singh',  'M', 16, '10A', 60, 58, 62),
('Divya Rao',     'F', 14, '9A',  88, 85, 80),
('Aditya Joshi',  'M', 15, '9B',  72, 68, 70),
('Meera Pillai',  'F', 16, '10B', 81, 89, 84),
('Rajesh Kumar',  'M', 17, '10A', 90, 87, 92),
('Isha Reddy',    'F', 15, '9A',  63, 66, 69);
```

Output after insertion — full table contents:

StudentID	Name	Gender	Age	Grade	MathScore	ScienceScore	EnglishScore
1	Aarav Sharma	M	15	9A	85	78	88
2	Priya Nair	F	16	10A	92	95	89
3	Rohan Mehta	M	15	9B	67	72	74
4	Sneha Kapoor	F	16	10B	78	82	91

StudentID	Name	Gender	Age	Grade	MathScore	ScienceScore	EnglishScore
5	Karan Verma	M	14	9A	55	60	65
6	Ananya Iyer	F	15	9B	95	91	93
7	Vikram Singh	M	16	10A	60	58	62
8	Divya Rao	F	14	9A	88	85	80
9	Aditya Joshi	M	15	9B	72	68	70
10	Meera Pillai	F	16	10B	81	89	84
11	Rajesh Kumar	M	17	10A	90	87	92
12	Isha Reddy	F	15	9A	63	66	69

12 rows inserted successfully with variety across gender (6 M / 6 F), age (14–17), grade (9A, 9B, 10A, 10B), and a realistic spread of scores (55–95) across all three subjects.

4. Queries, Output & Explanation

Query 1 — Show all student details

Code:

```
SELECT * FROM Students;
```

Output:

StudentID	Name	Gender	Age	Grade	MathScore	ScienceScore	EnglishScore
1	Aarav Sharma	M	15	9A	85	78	88
2	Priya Nair	F	16	10A	92	95	89
3	Rohan Mehta	M	15	9B	67	72	74
4	Sneha Kapoor	F	16	10B	78	82	91
5	Karan Verma	M	14	9A	55	60	65
6	Ananya Iyer	F	15	9B	95	91	93
7	Vikram Singh	M	16	10A	60	58	62
8	Divya Rao	F	14	9A	88	85	80
9	Aditya Joshi	M	15	9B	72	68	70
10	Meera Pillai	F	16	10B	81	89	84
11	Rajesh Kumar	M	17	10A	90	87	92
12	Isha Reddy	F	15	9A	63	66	69

Explanation:

Returns every column for all 12 students. This is the base dataset used for all further analysis below.

Query 2 — Average score in each subject

Code:

```
SELECT ROUND(AVG(MathScore),2) AS AvgMath,
       ROUND(AVG(ScienceScore),2) AS AvgScience,
       ROUND(AVG(EnglishScore),2) AS AvgEnglish
FROM Students;
```

Output:

AvgMath	AvgScience	AvgEnglish
77.17	77.58	79.75

Explanation:

The class average is highest in English (79.75), followed by Science (77.58) and Math (77.17). Math is the weakest subject on average, which could flag it as an area needing more teaching support.

Query 3 — Top performer (highest total score)

Code:

```
SELECT StudentID, Name, (MathScore + ScienceScore + EnglishScore) AS TotalScore
FROM Students
```

```
ORDER BY TotalScore DESC
LIMIT 1;
```

Output:

StudentID	Name	TotalScore
6	Ananya Iyer	279

Explanation:

Ananya Iyer is the top performer with a combined score of 279 out of a possible 300 across the three subjects.

Query 4 — Count students per grade

Code:

```
SELECT Grade, COUNT(*) AS NumStudents
FROM Students
GROUP BY Grade
ORDER BY Grade;
```

Output:

Grade	NumStudents
10A	3
10B	2
9A	4
9B	3

Explanation:

Grade 9A has the most students (4), followed by 9B and 10A (3 each), and 10B (2). This shows the class distribution is reasonably balanced across sections.

Query 5 — Average overall score by gender

Code:

```
SELECT Gender,
       ROUND(AVG((MathScore+ScienceScore+EnglishScore)/3.0),2) AS AvgOverallScore
FROM Students
GROUP BY Gender;
```

Output:

Gender	AvgOverallScore
F	83.94
M	72.39

Explanation:

Female students have a higher average overall score (83.94) compared to male students (72.39) in this sample. With only 12 records this is not statistically conclusive, but it illustrates how GROUP BY can reveal patterns worth investigating further with a larger dataset.

Query 6 — Students with Math score > 80

Code:

```
SELECT StudentID, Name, MathScore
FROM Students
WHERE MathScore > 80;
```

Output:

StudentID	Name	MathScore
1	Aarav Sharma	85
2	Priya Nair	92
6	Ananya Iyer	95
8	Divya Rao	88
10	Meera Pillai	81
11	Rajesh Kumar	90

Explanation:

Six students scored above 80 in Math: Aarav Sharma, Priya Nair, Ananya Iyer, Divya Rao, Meera Pillai, and Rajesh Kumar. These students could be considered for advanced math tracks or peer-tutoring roles.

Query 7 — Update a student's grade

Code:

```
UPDATE Students
SET Grade = '9B'
WHERE Name = 'Karan Verma';

-- Verification
SELECT StudentID, Name, Grade FROM Students WHERE Name = 'Karan Verma';
```

Output:

StudentID	Name	Grade
5	Karan Verma	9B

Explanation:

Karan Verma's grade was successfully changed from 9A to 9B. The verification query confirms the update was applied correctly, demonstrating that existing records can be safely modified after insertion.

5. Skills Demonstrated

• SQL database and table creation (DDL)

• Writing queries with SELECT, WHERE, GROUP BY

• Aggregation functions: AVG, COUNT, ranking with ORDER BY / LIMIT

• Filtering and updating data (WHERE, UPDATE)

• Interpreting query results like a data analyst

6. Deliverable

A .sql file containing the database/table creation, insert statements, and all required queries is provided alongside this report. All outputs shown above were generated by actually executing the queries against the dataset, not manually written.